Defense AI Observatory

An Underdeveloped Strategic Necessity

Defense AI in Pakistan

Umaima Ali

DAIO Study 25|28

Ein Projekt im Rahmen von
dtec.bw
Zentrum für Digitalisierungs- und Technologieforschung der Bundeswehr
Finanziert von der Europäischen Union
NextGenerationEU

About the Defense AI Observatory

The Defense AI Observatory (DAIO) at the Helmut Schmidt University in Hamburg monitors and analyzes the use of artificial intelligence by armed forces. DAIO comprises three interrelated work streams:

- Culture, concept development, and organizational transformation in the context of military innovation
- Current and future conflict pictures, conflict dynamics, and operational experience, especially related to the use of emerging technologies
- Defense industrial dynamics with a particular focus on the impact of emerging technologies on the nature and character of techno-industrial ecosystems

DAIO is an integral element of GhostPlay, a capability and technology development project for concept-driven and AI-enhanced defense decision-making in support of fast-paced defense operations. GhostPlay is funded by the Center for Digital and Technology Research of the German Bundeswehr (dtec.bw). dtec.bw is funded by the European Union – NextGenerationEU.

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Design

Almasy Information Design Thinking

Imprint

Umaima Ali, An Underdeveloped Strategic Necessity. Defence AI in Pakistan.
DAIO Study 25/28 (Hamburg: Defense AI Observatory, 2025).

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ISSN (online): 2749–5337
ISSN (print): 2749–5345

Content

1 Executive Summary

Pakistan's geostrategic context highlights its rivalry with India, territorial disputes over Kashmir, and internal threats from separatist movements and terrorism. Pakistan's defence strategy has traditionally emphasized conventional military strength, nuclear deterrence, and counterterrorism efforts. Pakistan's defence AI approach is primarily driven by the need to modernize its military in response to both kinetic and non-kinetic threats. Cybersecurity has become a national priority after multiple high-profile cyber incidents, prompting the government to invest in AI technologies for cyber defence, surveillance, unmanned systems, and precision effects. The launch of several key initiatives, such as the National Aerospace Science and Technology Park (NASTP) has catalysed Pakistan's drive to incorporate AI in the defence sector.

Pakistan collaborates with foreign partners, particularly China and Turkey, to develop its capabilities, with a focus on areas like unmanned aerial vehicles (UAVs) and missile systems. While these partnerships have advanced Pakistan's defence technology, concerns have emerged over data security and digital sovereignty. Additionally, Pakistan's nuclear deterrence strategy, which has traditionally relied on a balanced mix of conventional and nuclear forces, aims to incorporate AI for enhanced situational awareness, early warning systems, and decision support.

The Pakistan Air Force (PAF) and Pakistan Army have made significant strides in AI integration, especially with the development of advanced AI-powered drones, missiles, and cyber defence systems. Projects like the Shahpar III UAV and KaGeM V3 AI-guided cruise missile underscore the increasing role of AI in enhancing operational effectiveness. Furthermore, Pakistan is pursuing the development of a nuclear triad, with AI technologies being integrated into its strategic command systems for improved monitoring and control.

However, despite these advancements, Pakistan's defence AI efforts are still in the early stages, and gaps remain in its organizational and policy frameworks. The lack of a comprehensive national AI strategy has led to fragmented efforts across various branches of the military. The National Security Policy and the draft National AI Policy address AI's role in defence, but the primary focus remains on civilian applications, with some indirect implications for military modernization.

This study addresses the organizational challenges of adopting AI in defence, emphasizing the need for institutional reforms, interservice coordination, and dedicated units for AI-related projects. Pakistan's defence AI funding is gradually increasing, but it still lags behind global leaders. The budget allocation for research and development (R&D) remains limited, and the defence sector continues to rely heavily on imports to meet its technological needs. The establishment of centres like NASTP aims to bridge this gap by fostering local innovation, training AI talent, and enhancing collaboration with international partners.

The Ministry of Defence (MOD) plays a critical role in overseeing Pakistan's defence strategy and policy, managing the defence budget, and coordinating the activities of the three branches of the military—the Army, Navy, and Air Force. However, there is a notable challenge in achieving jointness among these services. While the MOD provides a centralized framework for defence management, the Army has historically dominated Pakistan's defence decision-making. Efforts have been made to foster better coordination, such as the establishment of the Joint Chiefs of Staff Committee (JCSC), but the services still operate largely independently in many areas, including AI development. Enhanced interservice cooperation will be crucial to ensuring the seamless integration of AI technologies and to effectively counter evolving threats in Multi-Domain Operations.

2 Thinking About Defence AI

2.1 Geostrategic Context

Since its inception in 1947, Pakistan's defence strategy has undergone significant change. Historically, Pakistan's strategic culture has reflected the country's history of betrayal, its fear of an irredentist Hindu India hostile to a Muslim Pakistan, and its conviction that the new state's identity should be rooted in its creation as a Muslim homeland. Pakistan's strategic culture is flexible and sensitive to political realism, and the Indian threat continues to shape it.[1]

Kashmir is a bone of contention between India and Pakistan and has grown to be a key component of Pakistan's defence policy. Pakistan and India had fought three wars over Kashmir: the 1947–48 war, the 1965 war, and the 1999 Kargil War.[2] The 1965 war was a pivotal event for Pakistan that emphasized the necessity of a strong defence strategy.[3] In 2025, India-Pakistan relations sharply escalated as a result of a deadly militant attack in Pahalgam, Indian-administered Kashmir.[4] The attack led to a low-intensity war between India and Pakistan, marking the biggest air battle in recent history.[5] Pakistan's current military strategy emphasizes counter-terrorism, force modernization, and nuclear deterrence. However, Pakistan has yet to institutionalize civil-military coordination in defence-making and issue a formal defence policy or white paper.[6]

Pakistan also faces internal threats from its neighbour due to India's backing of separatism and terrorism in provinces like Baluchistan.[7] However, India is not the only neighbour that threatens Pakistan internally and externally, the country has also endured contentious ties with Afghanistan. Pashtun separatism in the Khyber Pakhtunkhwa Province has been a significant internal threat for Pakistan. To preserve influence and security along the western border, the Pakistani government has traditionally backed pro-Pakistan groups in Afghanistan, such as the Taliban in the 1990s and the warlords in the 1980s.[8]

To counter internal and external threats, Pakistan has consistently sought foreign assistance for modern weapons and military training. During the 1950s, 1960s, and the 1980s, the U.S. was Pakistan's primary arms supplier.[9] In the 1950s, Pakistan aligned itself with the U.S. during the Cold War by entering Western

1 Briskey, "The Foundations of Pakistan's Strategic Culture."
2 "Kashmir Profile."
3 Aijaz, "Safeguarding the Future: Pakistan's Defence Policy and Strategy."
4 Center for Preventive Action, "Conflict between India and Pakistan."
5 El-Fekki, "India-Pakistan: 125 Jets Clash in One of Largest Dogfights in Recent History."
6 Aijaz, "Safeguarding the Future: Pakistan's Defence Policy and Strategy."
7 Lanlan/Fandi, "GT Investigates: Evidences, Sources Prove India 'Supports Terrorism' in Pakistan's Balochistan."
8 Lavoy, "Pakistan's Strategic Culture," pp. 13–14.
9 Ibid., p. 3.

defence pacts such as Central Treaty Organization (CENTO) and Southeast Asia Treaty Organization (SEATO) as part of its defence policy, which involved forming alliances to strengthen its military capabilities. Therefore, Pakistan had an effective deterrence against India. However, after the Indo-Pakistani War in 1965, the U.S. placed an embargo on military aid and stopped sending it to Pakistan.[10] Following the war, the U.S. wanted to prevent Pakistan from pursuing nuclear weapons. Subsequently, Pakistan turned to China and other countries to attain military technology.[11]

Traditionally, kinetic threats have been the major emphasis of Pakistan's national security strategy. But as offensive cyberattacks have become more frequent and intense, the focus has turned to non-kinetic, disruptive threats. For example, in 2018, a cyber-attack targeted Pakistan's banking system, compromising customer data and enabling fraudulent transactions. This incident exposed the vulnerabilities of financial infrastructure to cyber threats and spurred efforts to strengthen cybersecurity within the banking sector.[12] In 2021 another surge in spear-phishing email campaigns targeted Pakistan's Critical Information Infrastructures (CII) across both public and private sectors.[13] Furthermore, in July 2023, there was a cyberattack on the Election Commission of Pakistan (ECP), following which its authorities issued a security alert.[14]

Given the surge in cyber-attacks, Pakistan has started to focus on technological development. The Defence Production Policy of Pakistan recognizes critical and emerging defence technologies, analyses technological trends, identifies potential vulnerabilities, and ensures strategic advantage in key areas like artificial intelligence (AI), cyber warfare, and advanced weapon systems.[15] At the national level, the need to develop indigenous AI capabilities was first expressed by then-President Dr. Arif Alv, who launched the Presidential Initiative for Artificial Intelligence and Computing (PIAIC) to transform education, research, and business in Pakistan by using cutting-edge technologies.[16]

10 Aijaz, "Safeguarding the Future: Pakistan's Defence Policy and Strategy."
11 Lavoy, "Pakistan's Strategic Culture," p. 3.
12 Qarar, "'Almost All' Pakistani Banks Hacked in Security Breach, Says FIA Cybercrime Head."
13 These emails are deliberately titled to capture the recipient's attention, using subjects such as "List of Pak Defence Personnel," "DHA Multan Housing Scheme," "Pegasus Project Global Investigation," "Draft National Cyber Security Policy," and topics related to the Afghan situation. In some cases, phishing emails are even disguised with Pakistan Armed Forces logos to appear legitimate, increasing the risk of deception and security breaches. For more, see: NTISB Government of Pakistan, "Advisory - Cyber Attacks on Pakistan's Critical Information Infrastructure."
14 "Cyberattack on ECP; Security Alert Issued."
15 "Draft National Defence Production Policy."
16 "Presidential Initiative for Artificial Intelligence & Computing."

2.2 Defence AI Drivers

In Pakistan, defence AI is broadly understood as integrating AI into military decision-making, surveillance, logistics, autonomous systems, and cyber operations. According to an expert on emerging technologies in the defence sector, defence AI in Pakistan involves the use of autonomous systems, intelligence analysis, cybersecurity, decision support, surveillance, and predictive maintenance. Pakistan's rationale to use defence AI in Pakistan is to increase speed, efficiency, and strategic advantage in both combat and non-combat scenarios.[17]

While leading militaries have formalized frameworks for AI in Multi-Domain Operations (MDO), Pakistan is in the early stages of conceptual development.[18] Nonetheless, the country is exploring AI as a strategic enabler to support its core defence doctrines. Pakistan's defence strategy is based on four fundamental elements that work together to protect national security.

First, a robust defence against outside threats depends on military enhancement, modernization, and capabilities amplification.[19] In an era where global powers are rapidly advancing in defence AI, Pakistan recognizes that remaining passive would risk technological isolation. As a result, it also aims to adopt AI as a force multiplier to enhance its military capabilities and strategic posture.[20]

Second, a nuclear deterrent prevents aggression and preserves geopolitical stability by acting as a protective umbrella.[21] Pakistan has relied on both conventional military capabilities and nuclear deterrence for more than four decades to maintain competitive security and deter aggression from India. However, this balance is threatened by India's defence AI progress.[22]

Like other nuclear-armed states, Pakistan is unlikely to grant full autonomy to its nuclear platforms. However, AI and machine learning are being explored to enhance situational awareness regarding strategic assets, both during peacetime and crises. These technologies can support nuclear-related functions such as automatic target recognition (ATR), early warning systems, and decision support mechanisms. AI technologies are already being utilized by strategic organizations in combination with other technologies such as biometrics to advance the perimeter security of highly critical infrastructure.[23]

17 Interview with defence expert on emerging technologies, 26 June 2025.
18 Ilhsan, "Pakistan and the Rise of Artificial Intelligence in South Asian Warfare."
19 Aijaz, "Safeguarding the Future: Pakistan's Defence Policy and Strategy."
20 Interview with media expert, 27 June 2025.
21 Aijaz, "Safeguarding the Future: Pakistan's Defence Policy and Strategy."
22 Ihsan, "Pakistan's Use of AI Technology in Warfare."
23 Sial, "Military Applications of Artificial Intelligence in Pakistan and the Impact on Strategic Stability in South Asia."

Third, national security and anti-terrorism policies are essential for thwarting radical threats and safeguarding the state and its population.[24] In this context, AI can significantly bolster counterterrorism operations. By integrating AI into surveillance systems, security forces can improve their ability to monitor and interpret data using drones, ground sensors, and other sources.[25] If Pakistan incorporates drone technology and AI into its operations, its counterterrorism efforts could be enhanced and reinforced. For example, the Shahpar, a medium-altitude long-endurance drone, is perfect for surveillance and reconnaissance missions in hilly regions like Baluchistan. With AI-powered software, drones like Shahpar can improve their operational efficiency and targeting precision. In addition, AI can assist drones in differentiating between people and militants, thereby decreasing the likelihood of collateral damage and increasing strike precision.[26]

Fourth, strategic alliances and communication with China and other nations is meant to reduce the level of strategic risks Pakistan is facing, while at the same time enhancing the country's economic growth prospects.[27] The total arms imports from China between 2019 and 2023 alone were valued at $5.28 billion.[28] The cooperation between Pakistan and China is advancing in digital technology, AI, and information and communication technology (ICT) development.[29]

Overall, Pakistan's defence AI interest is fundamentally shaped by its strategic threat perception. To offset its strategic asymmetry with India, Pakistan views AI as a tool in modernizing its defence sector. The centre of Pakistan's defence policy, India, has integrated AI in its defence sector in many domains, including Intelligence, Surveillance and Reconnaissance (ISR), cyber security, decision support systems, battlefield planning and simulation, autonomous systems, and predictive maintenance.[30] India is strengthening its military posture as it increasingly relies on AI, particularly for offensive capability. India's growing AI reliance can change the balance of power in South Asia.[31]

Pakistan interprets India's defence AI developments as aggressive, triggering a reciprocal buildup. For example, in response to India's advanced drone capabilities, Pakistan chartered the use of drones for high altitude long endurance (HALE) "stealth UAS with strike capabilities" within its future unmanned aerial systems (UAS) vision.[32] In 2023, India deployed 140 AI-based surveillance systems along its

24 Aijaz, "Safeguarding the Future: Pakistan's Defence Policy and Strategy."
25 Javed, "AI-Driven Strategies to Counter Violent Terrorism and Extremism."
26 Hussain, "Why Drones and AI Must Lead Pakistan's Counter-Terrorism Strategy."
27 Aijaz, "Safeguarding the Future: Pakistan's Defence Policy and Strategy."
28 "China Supplying 81% of Pakistan's Military Imports: How Will Affect India-China Equation?"
29 Nazar, "Digital Technology, AI, ICT Development: Pakistan, China Agree to Strengthen Cooperation."
30 Mohan, "Passive Ambitions, Active Limitations: Defence AI in India", pp. 445–464; Khurshid and Zaman, "AI Integration in Indian Military: Appraisal of Security Apprehensions for South Asia," pp.1–15.
31 Ilhsan, "Pakistan and the Rise of Artificial Intelligence in South Asian Warfare."
32 Khan, "Is a Loyal Wingman UCAV on Pakistan's Procurement Roadmap?"

northwestern border with Pakistan.[33] Therefore, Pakistan's interest in using AI for defence reflects a threat-driven approach aimed at maintaining strategic stability. In addition, Pakistan also sees defence AI as a force and capability multiplier – the need for which stems from Pakistan's threat perception as well as the fear of missing out.

2.3 Capstone Documents

Pakistan has not officially confirmed using AI in its military weaponry and defences. The National Security Policy and the draft National AI Policy are the two documents that address the integration of AI in civilian and military affairs in Pakistan.

Pakistan's National Security Policy (NSP) 2022–2026, presented by the National Security Division (NSD), is the country's first formal national security document, outlining a citizen-centric approach within a "Comprehensive National Security framework." It acknowledges that traditional and non-traditional challenges and opportunities collectively impact national security. From a defence standpoint, the NSP highlights the importance of space-based technologies and technological enhancements to counter emerging security threats. Of particular interest is its focus on hybrid warfare—addressing disinformation, cyber threats, and economic coercion. While the document underlines the strategic role of technology in national security, it stops short of explicitly referencing AI.[34]

In contrast, the draft National AI Policy—developed by the Ministry of IT and Telecommunication (MIoTT) in 2023—is more expansive in its technological vision. Although primarily focused on economic transformation and social development, the policy sets the stage for defence AI adoption. Pakistan's National AI Policy aims to harness digital disruption to drive economic growth and improve the lives of its citizens. The policy focuses on several key areas, including establishing AI research and innovation centres to support the development, testing, and deployment of AI solutions. Additionally, awareness and readiness initiatives will equip Pakistan's workforce with the necessary skills to participate in the AI economy.[35]

The policy develops a structured roadmap to facilitate responsible and rapid adoption of AI. According to the draft policy, transformation and evolution efforts will integrate AI across industries through training programs and sectoral cooperation. R&D efforts focus on developing AI capabilities, standardizing data, and

33 Krishnan and Welle, "Indian Army Ramps up AI, but How Effective Will It Be?"
34 "National Security of Pakistan 2022–2026."
35 "Draft National Artificial Intelligence Policy."

fostering collaborations with both domestic and international institutions. Another critical aspect is the ethical and responsible use of AI, with the establishment of a regulatory body to ensure transparency, fairness, and accountability in AI applications (see also chapter 2.5).[36] Although defence-specific applications are not directly outlined, the policy implicitly supports military modernization by encouraging AI adoption through dual-use technologies. This suggests AI is largely conceptualized as a tool to enhance and modernize existing military practices rather than to initiate a disruptive overhaul of military power.

The National Defence Production Policy of Pakistan complements the above-mentioned policies by providing actionable goals to enhance national defence capabilities through technology. A central theme of the draft policy is the indigenization of defence technology and a strong push toward self-reliance. Establishing institutes and centres of excellence for specialized technologies is a top priority of this policy, with a focus on evaluating and implementing cutting-edge defence technologies such as sophisticated weapons, cyber warfare, and AI. The policy encourages local firms to participate in joint ventures and international partnerships to attain indigenous competence.[37]

To promote collaboration between public and private organizations in the defence production sector, public-private partnerships are encouraged. Defence production commercialization is viewed as a way to bring economic interests and national security together, although both must be properly balanced.[38] Through foreign partnerships, scholarships, and the merging of technical academic institutions, the policy emphasizes the necessity of producing top-notch human resources that meet international standards.[39] According to this policy draft, in all import indents involving large quantities and substantial capital, a minimum quota of 5–15% (of the total contract cost) should be kept for indigenous development using models such as licensed in-country manufacture, parallel industrial development, design house establishment, and local part manufacturing.[40] This policy reflects a systemic approach, which considers AI and other advanced technologies not in isolation, but as part of a broader defence-industrial strategy that aims to boost domestic innovation, strengthen strategic autonomy, and create a technology-driven security ecosystem.

36 Ibid
37 "Draft National Defence Production Policy," p. 12.
38 Ibid., p. 13.
39 Ibid., p. 12.
40 Ibid., p. 13.

2.4 Defence Data

Pakistan does not have a comprehensive national data strategy, but elements of such a strategy are evolving through sectoral policies, like the Digital Pakistan Policy (2018, revised in 2021) and the efforts under the Ministry of IT and Telecommunication.

While primarily civilian in orientation, this policy establishes the foundational digital infrastructure—such as AI labs, cloud systems, and data centres—that could potentially support future military applications. It also calls for comprehensive capacity-building initiatives—including awareness campaigns, trainings, seminars, and workshops—to expand the national talent pool and promote the adoption of smart technologies. Furthermore, the policy advocates for cost-effective integration of emerging technologies by encouraging open standards, providing targeted federal funding, and nurturing impactful public-private partnerships.[41] Although not explicitly framed as a defence initiative, the Digital Pakistan Policy contributes to the gradual evolution of a national data environment from which security and military sectors may also drive benefit.

The National Cyber Security Policy recognizes data sovereignty and infrastructure protection as national priorities. In Pakistan, only a limited number of Cyber Security Incident Response Teams (CSIRTs) are currently active across the public, private, and defence sectors. The state views any cyberattack on its critical infrastructure (CI) or critical information infrastructure (CII) as an act of aggression against national sovereignty and affirms its right to respond appropriately. According to the National Cyber Security Policy, Pakistan will operate in accordance with national and international cybersecurity frameworks, standards, and best practices, while expecting mutual respect for its digital sovereignty. To ensure effective implementation, a designated federal organization will be tasked with developing a comprehensive cybersecurity implementation framework.[42]

2.5 Regulation and Ethical Concerns

Pakistan is an outspoken supporter of regulating AI in the military domain. At the global level, Islamabad has also been advocating for a complete ban on lethal autonomous weapon systems (LAWS).[43] Concerns that AI could potentially be used in robots weaponized to kill or the delegation of life-and-death decisions

41 “Digital Pakistan Policy.”
42 “Government of Pakistan National Cyber Security Policy 2021.”
43 Ihsan, “Pakistan’s Use of AI Technology in Warfare.”

to machines are prompting Pakistan's stance on international regulation. In Pakistan's view, LAWS are immoral and – notwithstanding their level of sophistication – cannot be instructed to adhere to international humanitarian law (IHL).[44]

Pakistan has attended every Certain Conventional Weapons (CCW) meeting on autonomous weapons systems and is a signatory to the Convention on Certain Conventional Weapons. Pakistan also called for a legally binding global agreement to regulate AI-based weapon systems with respect to their development, testing, production, and deployment. The government has also supported a protocol to the Convention on Prohibitions or Restrictions on the Use of CCW that prohibits fully autonomous lethal armaments.[45]

Maintaining and enhancing human involvement in operational decision-making is central to Pakistan's position on AI. Pakistan became the first state in 2013 to openly call for a complete ban on LAWS at the meeting of UN Human Rights Council.[46] At the Sixth Review Conference of the CCW in December 2021, Pakistan stated that the "unique nature and characteristics' of autonomous weapons systems 'warrants further development of corresponding, legally binding international rules." In December 2023, Pakistan voted in support of UN General Assembly Resolution 78/241 on lethal autonomous weapons systems. In October 2024, Pakistan supported Draft Resolution L.77 on lethal autonomous weapons systems at the 79th UN General Assembly First Committee. In April 2024, Pakistan endorsed the Chair's Summary of the 2024 Vienna conference Humanity at the Crossroads: Autonomous Weapons Systems and the Challenge of Regulation.[47]

Pakistan's proposal for an international legal instrument on LAWS within the framework of the CCW advocates for a two-tier approach, combining prohibitions and restrictions to ensure compliance with IHL. The proposal defines LAWS as weapons that can autonomously select and apply lethal force without human intervention and calls for prohibitions of systems that lack human control. For systems not prohibited, the proposal includes restrictions such as limiting targets to military objectives, ensuring human intervention, and restricting operational scope.[48]

Pakistan demands that the Group of Governmental Experts in CCW creates an additional legally enforceable agreement that would ban some LAWS while regulating others. While resolving IHL's deficiencies as a regulatory framework, the supplementary protocol will fortify the international legal framework controlling LAWS.[49]

44 Fayyaz/Hussain/Andleeb, "Artificial Intelligence, Pakistan's National Security Policy: International Humanitarian Law (IHL)."
45 Ihsan, "Pakistan's Use of AI Technology in Warfare."
46 Butt, "AI-Driven Warfare: Navigating the Legal Ambiguities, Accountability Gaps, and Escalation Concerns."
47 "Pakistan."
48 Pakistan, "Elements of an international legal instrument on Lethal Autonomous Weapons Systems (LAWS)."
49 Saif-ul-Haq, "Governing Lethal Autonomous Weapon Systems (LAWS)- a Pakistani Perspective."

But despite the focus on international regulation of AI, Pakistan's AI policy lacks the detailed regulatory frameworks, for example, found in the EU's AI Act and Singapore's data protection laws. It fails to address critical issues such as data scraping for AI training, particularly when foreign companies are involved, and lacks provisions for cybersecurity and internet governance. The policy also does not mandate disclosure of AI deployment in sectors like marketing, social media, and political campaigns, risking privacy violations and misuse of AI.[50] Furthermore, the Islamabad Policy Research Institute (IPRI) has pointed out that using AI in counterterrorism operations also includes the potential for AI systems to make autonomous decisions in high-stakes situations, causing errors that could lead to civilian casualties or wrongful targeting. AI systems may also perpetuate bias, disproportionately affect marginalized communities, and violate privacy rights. IPRI advocates for a balance between national security and the protection of individual freedoms.[51]

Bearing these concerns in mind, the Centre for International Strategic Studies in Islamabad has argued that for Pakistan to integrate AI into its national security framework, it is essential to adopt a comprehensive approach that addresses ethical concerns, data privacy, and algorithmic biases. The think-tank advocates prioritizing human rights, accountability, and transparency. Additionally, strategies to mitigate biases in AI algorithms are crucial to ensure fairness and equity in decision-making processes.[52]

In sum, Pakistan is adhering to a human centric understanding of AI with machines supporting humans. Retired PAF Air Marshal Javaid Ahmed, among others, has raised concerns regarding AI replacing jobs. If AI has a disruptive potential, then the war on the control and proliferation of this technology also gains significance. Pakistani think-tankers are of the view that AI's ability to process vast amounts of data allows for first-shot capability and rapid adaptation in warfare, as seen in the use of AI in drones for ISR by Pakistan's own military. However, its application in warfare is debated, with some arguing AI could lead to completely autonomous wars, while others see its impact as evolutionary. AI in nuclear weapons control also raises concerns about destabilization.[53]

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50 Mahmood, "The Urgent Need for AI Regulation."
51 Hashmi, "The Ethical and Legal Challenges of Artificial Intelligence in Counter Terrorism Operations."
52 Baig, "Artificial Intelligence, Emerging Technologies and National Security of Pakistan."
53 "Artificial Intelligence (AI), Machine Learning (ML) and Implications for Pakistan on 20th August 2019."

3 Developing Defence AI

Defence AI in Pakistan remains largely exploratory, often linked to wider emerging technologies like automation, unmanned systems, cybersecurity, and decision-support systems. With no overarching policy or strategic vision to direct its advancement, Pakistan remains behind in the field of AI engineering. Nonetheless, two drivers shape defence AI development in Pakistan.

The first driver is India's defence AI advancement. Pakistan feels outperformed by India across all three main services (Army, Air Force, and Navy) as India is seen to quickly move forward by adopting new technology and working to improve its economic, social, and military position. Pakistan tries to catch-up but the national setup to do so is complex and challenging to manage. This also explains why Pakistan is prioritizing projects that refine current practices, rather than venturing into untested, disruptive AI paradigms.

In addition, Pakistan strives for technological sovereignty to reduce dependence on others. Digital or cyber sovereignty is of particular importance to Pakistan. It is understood as "establishing and maintaining the federal government's writ over content flows within the entire national cyber territory to protect core national interests and national values."[54] In this regard, Pakistan attaches great importance to building up and expanding indigenous digital and AI capabilities.

3.1 National Focus Areas and Key Actors

Defence Ecosystem

The MOD plays a key role in formulating and allocating the defence budget and overseeing military exercises involving all branches of the armed forces. Its departments include the Military Lands and Cantonments Department, Pakistan Civil Aviation Authority, Airports Security Force, and various intelligence bureaus. The Ministry also oversees civilian joint institutions such as the Defence and Strategic Studies Department at Quaid-i-Azam University. The Airports Security Force, established in 1976 and later placed under the MoD, ensures the protection of airports and civil aviation facilities in Pakistan.[55]

In general, Pakistan has a set up a broad network of research institutes that drive the push for indigenous technology development in traditional defence areas. In Pakistan, the Ministry of Defence Production (MoDP) is separate from its parent

54 Khalid, "Cyber Sovereignty in the Pakistani Context - Centre for Strategic and Contemporary Research."
55 "Ministry of Defence - Government of Pakistan."

agency, the MoD. MoDP is the primary institution responsible for overseeing and funding various defence-related organizations. These include entities like Pakistan Ordnance Factories (POF), Heavy Industries Taxila (HIT), Pakistan Aeronautical Complex (PAC), and others.[56] Pakistan's Defence Export Promotion Organization (DEPO) also operates under the MoDP and interacts closely with research and development (R&D) organizations and public and private manufacturers to facilitate foreign market access for a broad range of Pakistani defence products.[57]

The Research and Development Establishment (RDE), working under MoDP, is a defence-related R&D institute located in Rawalpindi, Punjab. The goal of the institution is to indigenize defence-related equipment requirements, vehicles, ammunition, bridges, products, trackways, weapons, optronics, and simulators. It aims to provide indigenous solutions to various end-users as well as friendly countries through partnerships with the public and private industry, academia and R&D setups within the country and abroad.[58]

Based in Rawalpindi, the Institute of Optronics (IOP), which was set up in 1984, is involved in producing and testing military-grade night vision equipment and image intensifier tubes.[59] Set up in 2000, the National Engineering and Scientific Commission (NESCOM) is in charge of Pakistan's major defence development initiatives, such as the Air Weapons Complex (AWC) and the National Defence Complex (NDC). The goal of NESCOM is to help Pakistan enhance its nuclear and ballistic missile capabilities while concentrating on conventional military equipment for export and domestic usage. It conducts research in several engineering fields including information technology.[60]

Overall, there are over 20 major public sector organizations and more than 100 private sector firms involved in manufacturing defence-related products, contributing to Pakistan's indigenous defence production. Among these entities Margalla Electronics is working on electronic equipment, radar systems, and communication equipment.[61] Global Industrial Defence Solutions (GIDS) is engaged in UAV platforms, flight control systems, C4I systems, datalinks, payloads, and ground support equipment. [62]

Against this background, several institutes have been established with the dedicated goal of advancing defence AI. The National Centre of Artificial Intelligence (NCAI) was established in 2018 at the National University of Science and Tech-

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56 Nazar, "Ministry of Defence Production: Government Decides to Turn 5 Entities into Autonomous Bodies."
57 "About DEPO."
58 "About Us."
59 Chandrashekhar, "Pakistan's Defence Industrial Base: An Overview."
60 Ibid.
61 Ibid.
62 Ibid.

nology (NUST) with the goal of creating AI technologies in partnership with the industrial and defence sectors.[63] As NCAI is responsible to facilitate AI innovation, research and advancement, it is expected that this organization will also help advance defence AI in Pakistan.[64] To encourage AI research in academia and industry, NCAI has allocated PKR200M (around USD704,000), giving priority to merit-based projects that tackle challenges in public health, agriculture, and neurology.[65] NCAI was launched under the government's initiative "Vision 2025." The project was launched with a funding of PKR1.1bn (around USD3.8M), which would be spent over three years to build and run labs at six of Pakistan's best universities.

Furthermore, several universities have established AI labs and units. The National University of Sciences and Technology (NUST) Islamabad has established a Deep Learning Lab and an Intelligent Field Robotics Lab; COMSATS Institute of Information Technology (CIIT) Islamabad launched a Medical Imaging and Diagnostics Lab; the NED University of Engineering & Technology Karachi established a Smart City Lab and a Neuro-Computation Lab; the University of Engineering and Technology (UET) Lahore established an Intelligent Criminology Lab; the Punjab University Lahore established an Agent-Based Modelling Lab; and the UET Peshawar established an Intelligent Information Processing and Intelligent System Design Lab.[66]

Complementing these efforts, the Pakistan Army, in 2022, launched the Army Centre of Emerging Technologies, dedicated to AI research in cybersecurity.[67] In addition, the Islamabad Policy Research Institute (IPRI) has argued that, at the strategic level, government agencies must develop military personnel's capacity by establishing the Centre for Army Artificial Intelligence and Computing.[68] This centre will be in charge of human-machine training so that operators and AI-enabled technologies may integrate seamlessly. Light-aid detachments from the Corps of Electrical and Mechanical Engineering (EME) should be included in every combat or support unit to provide maintenance training. The Pakistan Army's General Headquarters, the Ministry of Finance, and the Ministry of Defence should all contribute to the project's funding. In the end, the Army and PAF's centres can combine to form a tri-services facility.[69]

NASTP Alpha is a hub for cutting-edge research and development in AI and other emerging technologies. It hosts the PAF's largest organic R&D setups and provides

63 Rasheed, "How Pakistan Is Using Artificial Intelligence in Warfare Latest Military Technologies in 2025."
64 "AI in Pakistan's Defence Sector."
65 "Research Fund Projects."
66 Malik, Gillani, and Anwar, "Military's Guide to Artificial Intelligence."
67 Hussain/Khan, "South Asia's AI Arms Race."
68 Nizamani, "Use of Artificial Intelligence (AI) in Air Defence Systems and Unmanned Aerial Vehicle Drone Platforms: Identifying Areas of Improvement."
69 Ibid.

an attractive environment for aerospace, cyber, IT, fintech, and training companies. NASTP Alpha is also home to the National Incubator for Aerospace Technologies (NICAT), Siber Koza (a joint venture for international Cyber and IT incubation with CryptTech), and the NASTP Institute of Emerging Systems and Technologies (NIEST), which includes various training centres and schools. Additionally, it houses the National Centre for Cyber Security of Air University and the NASTP Aviation Academy. It has been approved as a Special Technology Zone (STZ) by the STZ Authority of Pakistan.[70]

NASTP Delta aims to cater to the needs of cyber, IT, computing, fintech, and training companies due to its proximity to prominent software houses. It plans to collaborate with top universities such as LUMS, University of the Punjab, and UET Lahore. NASTP Delta is focused on training the national youth through short-duration courses and certifications, especially for fresh graduates, in partnership with the Pakistan Software Export Board (PSEB), the MoITT, and the Pakistan Software Houses Association (P@SHA), a non-profit trade association. The IT Park at NASTP Delta is a collaborative facility for academic institutions, high-tech companies, entrepreneurs, and startups to share knowledge, innovate, and work on joint projects. NASTP Institute of Information Technology (NIIT), located within NASTP Delta, provides industry-relevant certifications, acting as an incubator for students to connect with the corporate world. Affiliated with the Air University, NIIT offers programs that support startups and transform students into skilled human resources for the IT industry.[71]

NASTP Silicon 1 is home to the major design and R&D setups of the PAF in Simulator, AR, VR, Robotics, Space, Wireless and Communication Systems, Cyber, IT, Computing & Big Data, Software Development, Artificial Intelligence domains.[72] NSTAP allows the defence sector to cooperate with the private sector. However, the leader to develop defence AI in Pakistan is the defence sector. For example, the private companies working in NASTP are guided and led by the PAF.[73]

Non-Defence Ecosystem

In addition, Pakistan has set up several institutions to generate and advance knowledge about AI in general. This step is much needed as the technological divide between Pakistan and other nations is widening due to Pakistan's poor AI curriculum, lack of data, AI specialists, and domestic infrastructure.[74]

70 "NASTP | Alpha Rawalpindi."
71 "NASTP | Delta Lahore."
72 "NASTP | Silicon 1 Karachi."
73 Interview with NASTP expert, 27 June 2025.
74 Salamat, "AI in Military Security."

In 2009, the Institute of Business Administration (IBA) has established its AI Lab with the goal of providing a platform for the creation of intelligent systems and stimulating conversations about new developments in the field of AI.[75] The fact that several other top universities, including the National University of Computer and Emerging Sciences (NUCES, Air University), and the Centre for Advanced Studies in Engineering (CASE), among others, have begun to offer bachelor's degrees in AI is also a positive development in the field. It is a much-needed step in the right direction because it will help develop future AI-specialized indigenous human resources.[76]

The Research Centre for Artificial Intelligence (RCAI) was founded in 2016 by the NED University in Karachi to advance theoretical and applied knowledge of AI in academia and business as well as to support the planning and development of policies. The centre wants to support and advance AI in Pakistan. Along with maximizing Pakistan's involvement in the global AI community, it also seeks to offer training grounds at different academic levels and consulting services to the government and industry. The research centre focuses on application areas in the domains of water and food resources, finance, energy, smart grid, cybersecurity, robotics, transportation, decision support systems, and building infrastructure. Additionally, this enables RCAI to develop hardware and software-based integrated solutions.[77]

Furthermore, there is the National Centre of Robotics and Automation, which consists of eleven labs at thirteen university in Pakistan. Its mission is to build a national technology ecosystem and help solve local problems.[78] In 2018, the centre received USD1.67M for AI projects in the fields of robot design and development, industrial monitoring and automation, and human-centred robotics.[79]

Overall, Pakistan achieved a major milestone in 2023, when the Prime Minister unveiled a National Action Plan for AI integration, which included the creation of a specific organization to supervise the growth of knowledge about AI-based systems. As part of this effort, Pakistan established a task force to hasten the deployment of AI and set the lofty goal of training one million people in AI-related skills.[80]

75 Malik/Gillani/Anwar, "Military's Guide to Artificial Intelligence."
76 Ibid.
77 Ibid.
78 "National Centre of Robotics and Automation."
79 Hussain/Khan, "South Asia's AI Arms Race."
80 "National Taskforce Sets Roadmap for AI-Powered Sectoral Transformation, National Development."

3.2 Nuclear Forces

Pakistan maintains an assertive nuclear command-and-control structure, with the Prime Minister heading the National Command Authority (NCA) and the Strategic Plans Division (SPD) serving as its secretariat.[81] To reinforce central oversight, strategic force commanders have custody of delivery systems, while nuclear warheads remain under the direct control of the SPD, ensuring a clear separation. Pakistan has cultivated a robust security culture surrounding its nuclear assets. A key element of this is the geographical separation between warheads and delivery systems, which helps prevent accidental or unauthorized launch. However, Pakistan's sea-based deterrence marks a shift toward a more ready arsenal, as warhead-delivery separation becomes technically unfeasible at sea, shortening launch times and increasing alert status.[82]

The survivability of nuclear forces is a major concern in nuclear South Asia. There is low chance that Pakistan would give full autonomy to its nuclear platforms. However, AI and machine learning are increasingly being integrated to enhance decision-making and strategic awareness. AI can enhance situational awareness through automatic target recognition (ATR), early warning systems, and decision support tools.[83] To this end, Pakistan has established a National Command Centre (NCC) equipped with a fully automated Strategic Command and Control Support System (SCCSS). This system provides real-time monitoring and situational awareness of all strategic assets, ensuring centralized oversight during any operational deployment.[84]

Pakistan keeps adding to its nuclear arsenal and developing its delivery systems. According to a SIPRI report, Pakistan is developing a sea-based nuclear force to create a nuclear triad and a credible second-strike capability. At the heart of this effort is the Babur-3 submarine-launched cruise missile (SLCM). Babur-3 is intended to give nuclear capability to the Hashmat-class diesel-electric submarines of the Pakistan Navy.[85] Pakistan is investing in its subsea-based nuclear deterrent; however, the effectiveness of submarine stealth has been reduced due to the advances in detection technologies. AI-enabled systems that analyse sensor data can easily spot subtle anomalies in the ocean, such as changes caused by a passing submarine.[86] The Indian Navy is expanding an advanced underwater sensor network across key areas of the Indian Ocean Region (IOR) to detect and track the growing submarine activity of the Chinese by using cutting-edge technology for early detection and real-time tracking.[87] Pakistan's pursuit of a nuclear triad still relies

81 Afeen, "Pakistan's Nuclear Command and Control Structure: Ensuring Security and Deterrence."
82 Siddiqui, "Pakistan Army Conducts Successful Test Launch of Surface-To-Surface Babur Cruise Missile."
83 "Military Application of Artificial Intelligence in South Asia on 14th May 2020."
84 Ahmed, "Pakistan's Tactical Nuclear Weapons and Their Impact on Stability."
85 Kristensen/Korda, "Pakistani nuclear forces."
86 Bajema, "Will Even the Most Advanced Subs Have Nowhere to Hide?"
87 "India to Launch Advanced Underwater Sensor Network in Indian Ocean to Counter China."

on conventional submarine platforms and the escalating regional Anti-Submarine Warfare (ASW) capabilities expose critical limitations. However, If Pakistan is advancing its Babur-3 capabilities, it is plausible that future iterations could incorporate AI for autonomous guidance, although this has not been officially confirmed.

3.3 Air Force

In 2020, the Pakistan Air Force (PAF) launched a course on cognitive electronic warfare and created the Centre for Artificial Intelligence and Computing (CENTAIC). The PAF will be able to incorporate AI into its operational domain due to its role as the nation's leader in AI development for both military and civilian applications.[88] In this regard, CENTAIC will play a key role in collaborating with industry, academia and leading institutions like the National Science & Technology Park (NASTP) and the Centre for Aerospace and Security Studies (CASS).[89] NASTP creates opportunities for joint work between industry, academia and the government, turning it into a technology ecosystem in the country.

AI and machine learning technologies are necessary for quick analysis and quick decision-making on the battlefield due to the massive amounts of data generated by modern linked weapon systems. Pravin Sawhney, an Indian defence analyst, thinks the PAF employed Cognitive Electronic Warfare (CEW) in Pakistan's successful Operation Swift Report, which was initiated in reaction to India's 2019 bombing of Balakot. According to Sawhney, the Pakistani military has realized the value of deploying its air force as the lead branch for deploying AI/ML and CEW following the success of Operation Swift Retort.[90]

Currently, the PAF uses many different UAVs, the JF-17, improved F-16s with Mid-Life Updates, and the J-10CE. Additionally, plans are in motion to incorporate stealth platforms such as the J-31. Strengthening combat readiness, maintaining air superiority, and offering a credible deterrent to address both domestic and regional security issues have been the strategic goals of recent acquisitions and system improvements. The PAF now can use UAVs for both strike and surveillance missions. To improve decision-making, R&D efforts focus on incorporating AI into its operational systems. Although significant advancements remain to be achieved, this approach underlines Pakistan's willingness to augment existing legacy systems with AI.[91]

88 Mailk/Gillani/Anwar, "Military's Guide to Artificial Intelligence," pp.25.
89 Khan, "Pakistan Air Force's NASTP Working on Indigenous AEW&C System."
90 Haq, "Pakistani Military Launches Defence AI Program."
91 Mengal/Mustafa, "Emerging Technologies and its impact on Air Power."

The development of the Shahpar III medium-altitude long-endurance UAV is considered as an important achievement of the PAF.[92] Shahpar III uses AI for surveillance, target acquisition and precision strikes.[93] In addition, the KaGeM V3, a smart air-launched cruise missile, uses AI-powered guidance systems.[94]

Historically, Pakistan has used human sensors, aircraft early warning systems, and ground-based radars and sensors to detect and identify threats to its airspace. This approach presents challenges as interoperability of software codes and interfaces need to be advanced to provide for integrated air defence, for example. Hardware manufacturers typically protect the software codes of their platforms from end users due to intellectual property concerns. This makes it more difficult to integrate systems under a single operating system. However, data mining and sensor fusion are prerequisites for integrating AI into air defence platforms.

The PAF reportedly stores vast amounts of data from antiquated platforms including Air Defence radars on magnetic tapes. PAF is probably using cloud computing services to store, archive, and exploit this data for training-related objectives. Even if such data might not be totally helpful for detecting contemporary platforms, processed data can still be used to train algorithms through repeated simulations for AI-enabled Air Defence platforms to detect and identify current airborne threats. In particular, it can facilitate the development of predictive modelling capabilities in AI-powered Air Defence systems, which can project adversary platforms' flight paths, altitudes, and potential movements in real-time scenarios.[95] The use of unmanned fighter jets, or drones, is part of the "Loyal Wingman" philosophy.[96]

3.4 Cyber Forces

In a webinar organized by ISSI, a speaker categorized cyberattacks targeting Pakistan into three main groups: attacks on individuals, organizations, and government institutions. A key issue, according to him, is Pakistan's heavy reliance on third-party solutions instead of developing indigenous cybersecurity frameworks.[97] Pakistan is committed to protecting its critical assets and public, as reflected in its National Security Policy and National Cybersecurity Policy. To enhance response effectiveness, the government must first identify critical assets and perform com-

92 "GIDS - Products."
93 Rasheed, "How Pakistan Is Using Artificial Intelligence in Warfare Latest Military Technologies in 2025."
94 Ibid.
95 Nizamani, "Use of Artificial Intelligence (AI) in Air Defence Systems and Unmanned Aerial Vehicle Drone Platforms: Identifying Areas of Improvement."
96 Fayyaz/Hussain/Andleeb, "Artificial Intelligence, Pakistan's National Security Policy: International Humanitarian Law (IHL)."
97 Mustafa, "ACDC Special Report: Comprehensive National Security and Emerging Technologies."

prehensive risk assessments. A layered defence strategy and increased awareness within organizations can help prevent significant damage.[98] So far, however, official Pakistani policy documents do not mention using AI for cyber defence.

3.5 International Partners

Currently, Pakistan's defence industry is largely limited to the production of ammunition. The country is still either an importer or does offset and licensed production, with limited technology transfer. With China's help, Pakistan manufactures the JF-17 Thunder fighter aircraft. In the past, Pakistan used to rely on Russia for engines; today the country relies on China for imported parts. Despite the development of many UAVs at home (such as Uqab, Shahpar, Salar and Burraq), Pakistan has purchased Bayraktar TB2 and Akinci attack drones from Turkey. Pakistan imports Turkish TB2 drones equipped with advanced AI[99] and advances the proficiency of drone operators via joint training with Baykar.[100] Pakistan relies heavily on Turkey and China in the maritime domain also; the production of frigates, submarines, or other military supply vessels is being done in Pakistan with the help of Turkey and China.[101]

In 2021, Pakistan's National Engineering and Science Commission signed an agreement with Turkish Aerospace Industries to co-produce Anka military drones in Pakistan.[102] In addition, the PAF is equipped with the Anka UAVs and Bayraktar AKINCI, TB2, and TB3 UAVs, which are all capable of launching missiles. The Kemanke cruise missile has AI built into it, allowing it to find and select targets and choose them on its own even in bad weather.[103]

Pakistan and Turkey are expanding technology collaboration. In a meeting between Pakistan's Ministry of Information Technology and Telecommunications (MoITT) and Turkey's ambassador in 2025, Turkey's ambassador highlighted Turkey's advancements in digital transformation, including e-governance, AI, and 5G. The countries agreed to foster business-to-business cooperation, particularly between Pakistan's National Information Technology Board (NITB) and Turkey's TurkSat.[104]

98 Ibid.
99 "Powered by Artificial Intelligence and a Turbo Engine."
100 Mehdi, "Pakistan's Acquisition of Turkish Drones and India's Options."
101 "Pakistan's Defence Industries; Obstacles and Future Prospect."
102 Andreopoulos, "Turkey and Drone Warfare in the Pakistan-India Conflict."
103 Sheikh, "The New Era of Pakistan-Turkey Defence Ties."
104 "Pakistan and Turkey Discuss IT Cooperation."

Pakistan heavily relies on defence imports, with China emerging as its primary partner in recent years. China boasts the largest military budget in the region, at USD232bn in 2024, and is Pakistan's main supplier. Pakistan is ranked as the world's fifth-largest arms importer. Between 2019 and 2023, it sourced 82% of its arms imports from China.[105] Thanks to cooperation with China, Pakistan can also expedite developing AI. The creation of the China-Pakistan Intelligent Systems (CPInS) Lab in 2022 marked the beginning of cooperation in this area. In addition, the Pakistani government strived to expedite the rollout of fifth generation (5G) in the nation by August 2024. This would have enabled real-time data analysis by enabling 5G cloud computing and telecoms to effectively transmit, gather, and process massive volumes of combat data.[106] However, the country faced the biggest global economic losses from internet outages in 2024, totalling USD1.62bn, while still wating for the 5G rollout in January 2025.[107]

Chinese investments have enabled Pakistan to enhance its AI capabilities, helping reduce the technological gap with other global players. However, this partnership raises concerns regarding data security and privacy. China's dominance in AI gives it leverage over Pakistan's digital infrastructure, which could compromise Pakistan's digital sovereignty. The reliance on Chinese AI platforms and systems raises concerns over potential surveillance, data theft, and the loss of control over critical digital systems.

105 Palve, "With Stealth Jets, AIP-Subs, SAMs, UAVs on Radar, Pakistan Makes Audacious Hike in Defence Budget."
106 Babar, "Unlocking IoMT and AI's Potential for Pakistan's Defence Infrastructure."
107 "IT experts call on govt for urgent 5G rollout to combat slow internet."

4 Organizing Defence AI

As Pakistan is warming up to the use of AI in defence and beyond, the existing national security setup is very likely to produce huge challenges. Institutional complexity, the lack of clear strategic guidance, and decades of inter-service and inter-organization rivalry do not bode well for the need to advance digitalization and create an environment, which is amenable to cross-organizational data sharing. Beyond national security and defence, however, things look slightly better with different initiatives launched to improve government services.

Pakistan's national security architecture has evolved gradually. The Cabinet Committee for Defence (DCC), initially part of Pakistan's defence decision-making structure, was limited in its authority. This allowed the service chiefs to exercise significant independence. Additionally, the lack of joint planning among the military services exacerbated coordination issues, with the Army, due to its size and political influence, dominating the other services. In 1973, the Bhutto government introduced reforms that placed national defence responsibility under the Prime Minister and restructured the military command. The DCC, chaired by the Prime Minister, became the central body for defence decision-making, supported by a multi-tiered bureaucratic structure.[108]

In Pakistan, DCC was responsible for determining the size, role, and shape of the armed forces.[109] However, in 2013, DCC was reframed as the Cabinet Committee on National Security (CCNS). CCNS serves as the highest institutional forum where federal ministries, civilian intelligence agencies, and military institutions collaborate under the guidance of political leadership to formulate policies. The goal of CCNS was to develop a national security policy, which would serve as the framework defence, and other national security-related policies. Additionally, a National Security Division (NSD) was established within the federal civil service to support the CCNS, supervised by a senior bureaucrat.[110]

The Ministry of Defence (MOD) is also theoretically significant, tasked with handling policy and administrative matters related to the armed forces. According to the 1973 rules of business, the Army, Navy, and Air Force are required to route their official matters through the MOD. Despite formal lines of authority, the appointments, budgets, and operations of Pakistan's key military intelligence agencies—the Inter-Services Intelligence (ISI) and Military Intelligence (MI)—are managed by the Army's General Headquarters, rather than the civilian executive.[111]
In addition, the MoD has not been successful in resolving inter-service rivalry, a common issue in Pakistani power politics. Its bureaucratic structure, which includes

108 Siddiqa-Agha, Pakistan's Arms Procurement and Military Buildup, pp. 37–38.
109 Pattanaik, "Civil-Military Coordination and Defence Decision-Making in Pakistan."
110 Rumi, "Charting Pakistan's Internal Security Policy."
111 Ibid.

a large workforce of civil servants and military personnel, is divided into multiple divisions—particularly the Defence and Defence Production divisions.[112]

Today, the Strategic Plans Division (SPD), a military organization under the Joint Chiefs of Staff Committee (JCSC), likely oversees defence data.[113] The SPD functions as the Secretariat for the National Command Authority (NCA) and perform functions relating to planning, coordination, and establishment of a reliable and effective Command, Control, Communications, Computers, Intelligence and Interoperability (C4I2) network.[114] The Chairman of the JCSC, a four-star general, coordinates with all services and assists the Prime Minister in wartime. However, the committee cannot directly command troops or interfere with the day-to-day operations of the armed forces, particularly when the Army is in power, which has hindered its effectiveness. Despite these organizational efforts, the system remains fraught with inefficiencies, with the military services often operating independently rather than as a cohesive unit.[115] In addition, JCSC has often been a passive institution, with the military services making independent decisions. Despite efforts to maintain civilian supremacy, the military has often played a dominant role in defence decisions.[116]

Historically, the defence decision-making process in Pakistan has long been influenced by a strong pro-military lobby, with the Army holding more sway over policy matters than the Navy or Air Force. The Army's dominance is a result of its larger size, its central role in the country's political landscape, and its historical influence over national governance. During the creation of the JCSC in the 1970s, the Army's dominance was clear, with the Army chief often holding dual roles, such as head of the military and President of the country. [117]

While the PAF is smaller compared to the Army, it plays a crucial role in the country's defence strategy, particularly due to its need for advanced technology and its capital-intensive nature. On the other hand, the Pakistan Navy has historically been the most neglected service, largely due to its lack of political influence and its secondary role in Pakistan's defence strategy, which has traditionally focused on land warfare. Both the Air Force and Navy face challenges in acquiring the resources they need due to the Army's dominant influence in defence decision-making. Their procurement strategies often rely on the personalities of their service chiefs and their relationships with the Army and civilian leadership, highlighting the power dynamics within Pakistan's military establishment.[118]

112 Siddiqa-Agha, Pakistan's Arms Procurement and Military Buildup, p. 46.
113 Interview with media expert, 27 June 2025.
114 "Strategic Plans Division (SPD)."
115 Siddiqa-Agha, Pakistan's Arms Procurement and Military Buildup, pp. 46–49.
116 Pattanaik, "Civil-Military Coordination and Defence Decision-Making in Pakistan."
117 Siddiqa-Agha, Pakistan's Arms Procurement and Military Buildup, p. 63–77.
118 Ibid.

The Military Finance wing in the Ministry of Finance plays a crucial role in managing the financial aspects of defence operations. It monitors funds allocated for military purposes, working closely with the MoD. The Military Accountant General (MAG) also oversees payments and audits military expenditures, but its reliance on the defence budget and lack of independence from the defence establishment has raised concerns about the transparency and accuracy of its financial oversight. The Defence Production Division manages both foreign and domestic weapon acquisitions, with a structure that includes smaller procurement offices and large facilities like the Pakistan Ordnance Factories (POFs) and Pakistan Aeronautical Complex (PAC). Despite its hierarchical structure, the Army has significant influence in this system, particularly in the procurement of major weaponry, due to its dominant position within the military.[119]

On the non-military side, the Pakistan Security Standard for Cryptographic and IT Security Devices (PSS), developed by the National Telecom & Information Technology Security Board (NTISB), outlines a national framework for evaluating, certifying, and securing information technology and cryptographic equipment. The PSS framework promotes technological self-reliance, regulatory oversight, and international alignment in cyber and information security. It covers certification, auditing, development, and export of secure technologies to safeguard critical information assets across sectors.[120]

Beyond national security and defence, the MoITT, alongside other stakeholders, is committed to making Pakistan a global digital economy leader, with universal access to digital services. The recently enacted Digital National Pakistan Act (DNP) 2025 lays the foundation for a digital society, economy, and governance, supported by strong digital infrastructure. A central part of Pakistan's approach is the creation of a National Digital Commission (NDC) and a Pakistan Digital Authority (PDA), which will guide the nation's digital transformation, oversee AI adoption, and ensure secure implementation of emerging technologies like AI, cloud computing, and virtual assets. Pakistan also plans to contribute to global initiatives like the Partnership on AI to Benefit People and Society, which focuses on the ethical use of AI.

In 2025, the Ministry of Planning, Development, and Special Initiatives (MoPDSI) convened a key meeting of the National AI Taskforce, chaired by Minister Ahsan Iqbal, to discuss the AI Implementation Roadmap. This initiative aims to enhance Pakistan's productivity, innovation, and global competitiveness by integrating AI across various sectors. A central part of the plan includes the Uraan Pakistan initiative, which aims to drive export growth, digital transformation, sustainable

119 Ibid., p. 46.
120 "Pakistan Security Standard for Cryptographic & Information Technology Security Devices."

development, energy efficiency, and social justice. The AI Implementation Plan focuses on AI education, skill development, research, and the creation of Technology Parks. A National AI Office will oversee AI governance and regulation, and regional AI Liaison Offices will address localized needs. The plan also includes global partnerships for knowledge exchange, AI tech parks to foster innovation, and a focus on building AI infrastructure and a skilled workforce. [121]

Pakistan has launched an AI-based customs clearance and risk management system developed by the Federal Board of Revenue (FBR) to enhance transparency, reduce human intervention, and improve cross-border trade efficiency. The new system utilizes AI, bots, and machine learning to automate the classification and valuation of goods. The aim is to reduce delays, streamline tax assessment, and minimize manual interference. Initial testing showed a 92% improvement in performance and an 83% increase in identified goods declarations for tax collection, along with a significant rise in green channel clearances. The government also plans to use video analytics to improve tax collection in the manufacturing sector, with early testing showing 98% efficiency. The new system addresses long-standing issues in Pakistan's customs process, including delays, inefficiencies, and corruption, by reducing reliance on manual procedures and improving risk profiling.[122]

In addition, the government has launched an AI curriculum across federal schools. Training 10,000 AI instructors has started, and the government has also initiated fibre optic internet connectivity for 532 schools in Islamabad. These efforts were made as part of Prime Minister Shehbaz Sharif's "Digital Nation Pakistan" vision. More than 1,000 teachers have already been trained. According to the initiative, starting with 2025, AI and emerging technology education will now be made accessible to every child.[123]

121 "AI-Powered Future: National Taskforce Sets Roadmap for Sectoral Transformation and National Development."
122 Shabbir, "Pakistan Launches First-Ever AI Customs System to Boost Transparency, Efficiency."
123 Nazar, "'Digital Nation Pakistan': AI Curriculum across Federal Schools Launched."

5 Funding Defence AI

Whether Pakistan's government is civilian-led or under military rule (which has been the case for most of its history), defence has always been its top budgetary priority – but defence R&D only plays a minor role. Specific spending figures for defence AI are not publicly available.

Alongside its nuclear capabilities, Pakistan relies on a strong conventional military force.[124] In 2025, the government endorsed an 18% increase in defence spending due to tensions with India.[125] Pakistani defence spending decreased from around 2.6% of the Gross Domestic Product (GDP) in 2020 to 1.7% in 2024.[126] Accordingly, the revised budget for defence in 2023–2024 was USD6.489bn, while in 2024–2025, it was USD7.427bn.[127] Out of this amount USD2.853bn were allocated to employee related expenses, USD1.797bn to operating expense, USD1.92bn to physical assets and USD857.06M to civil works.[128]

The FY 2025–2026 budget provision for defence R&D was USD6.9M compared to USD9.7M FY 2024–2025. In addition, for FY 2024–2025, the budget for administrative support to the Defence Forces and attached civil departments/policy making and coordination was USD7.3M. The budget provision for development schemes (universities, educational institutes, cantonment R&D complex) was USD8,8M for the same year.[129]

On the other side, Information Technology and Telecommunication Division got zero funds to develop the human capital to utilize its true potential for the uplift of the sector for FY 2024–2025 and USD7M ICT and Research & Development Fund for FY 2025–2026.[130]

Key drivers of the budget include dispute with India, separatist movements in the province of Baluchistan, India's revocation of Article 370 and the threat of terrorism.[131] The main budget spending categories include naval vessels and surface combatants, missiles and missile defence systems, military fixed wing aircraft, military land vehicles, ammunition, submarines, and artillery systems.[132]

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124 Lavoy, "Pakistan's Strategic Culture," p.3.
125 "Pakistan to Increase Defence Spending by 18% in Budget: Report."
126 Palve, "With Stealth Jets, AIP-Subs, SAMs, UAVs on Radar, Pakistan Makes Audacious Hike in Defence Budget."
127 "Federal Budget 2024–2025," p. 10.
128 Ibid., p. 18.
129 "Federal Budget 2025–2026," p. 49.
130 Ibid., pp.127–128.
131 In August 2019, Articles 370 and 35A of the Indian Constitution, which granted Kashmir special status, were scrapped unilaterally by the Indian government. The amendment changed the rights of landowners and gave state residents preference in employment and education.
132 Globaldata, "Pakistan Defence Market Size and Trends, Budget Allocation, Regulations, Key Acquisitions, Competitive Landscape and Forecast, 2024–2029."

Historically, the average R&D expenditure value for Pakistan from 1997 to 2021 used to be 0.25%. In 2021, it was 0.16% as opposed to the global average value of 1.25%.[133] Today, only 0.2% of Pakistan's industrial sector GDP is devoted to R&D.

The Centre of Excellence in Artificial Intelligence (CoE-AI) will act as a focal point for AI-related research and development in both the public and private sectors to close this gap and foster innovation. To finance R&D initiatives aimed at addressing socioeconomic, civic, and behavioural issues, a special grant program will be created. To help them go from the prototype stage to commercial viability and investment readiness, up to 400 ideas will receive financial and logistical support of up to USD3,500 each, or another sum depending on certain requirements. Furthermore, the CoE-AI will fund scholarly investigations and theses that investigate cutting-edge theories and models in AI and other fields.[134]

Currently, funding for AI/ML/DL-led R&D initiatives in academia and industry remains limited through the available public-sector funding channels.[135] Pakistani technology startups leveraging AI-based solutions secured USD278M in funding in 2021.[136] In addition, defence research focused institutions like NASTP also receive seed funding to support their initiatives. For example, in 2022, four startups of NASTP won seed funding of USD35,000 through the Innovator Seed Fund (ISF), an initiative under Pakistan's Higher Education Commission's Higher Education Development Program (HEDP) for engaging entrepreneurs who want to convert their business ideas into action.[137]

133 "Pakistan: Research and Development Expenditure."
134 "Draft National Artificial Intelligence Policy."
135 Ibid.
136 "2021: A Banner Year for Pakistani Tech Startup Investments."
137 "Four NSTP-Based Startups Win $35,000 Seed Funding by Innovator Seed Fund (ISF)."

6 Fielding and Operating Defence AI

Currently, Pakistan is focusing on using AI to improve enhanced situational awareness, advance combat effectiveness, improving precision weapons, and bolstering cyber defence. Several projects have been launched in each of these four areas:

- **Advancing Combat Effectiveness**
  The primary goal of using AI is to advance force multiplication in combination with unmanned systems. One of the most notable achievements of the PAF is the development of the Shahpar III. The SHAHPAR-III is a Medium Altitude Long Endurance (MALE) Unmanned Combat Aerial Vehicle (UCAV) designed to deliver superior performance in surveillance, reconnaissance, and combat missions. Its combination of advanced features, robust design, and versatile capabilities make it an indispensable force multiplier for modern military operations.[138] Developed by National Engineering and Scientific Commission (NESCOM), a Pakistani government-funded defence contractor, and marketed by Global Industrial & Defence Solutions (GIDS) Shahpar III uses AI for surveillance target acquisition and precision strikes.[139]

- **Improving Precision Weapons**
  Here the main purpose of using AI is to improve targeting accuracy and reduce collateral damage by using AI-guided weapons. CENTAIC is exploring important areas of AI such as big data, machine learning, deep learning, predictive analytics and natural language processing. The institute is also running Azm, a cutting-edge weapons program. It is assumed that AI may be used in this project to improve human-machine interfaces (HMI), advance image processing for IR seekers, and develop algorithms for guidance systems used in air-to-air and air-to-surface missiles.[140]

  KaGeM V3, an intelligent air-launched cruise missile using an AI-enhanced guiding system, is another PAF product. Developed in cooperation with Turkey, the AI enhancement improves mission flexibility and targeting precision. Additionally, Pakistan has created the Spider system, an electronic warfare tool that combats drones. This transportable device provides a protective barrier against aerial threats by using AI to identify, track, and jam hostile UAVs. An AI-powered laser defence robot was recently created by Mehran University of Engineering and Technology (MUET) students for the civilian market. It shows how Pakistan's youth is advancing national security innovation with this robot, which uses autonomous logic to detect, track, and eliminate impending threats.[141]

138 "GIDS - Products."
139 Rasheed, "How Pakistan Is Using Artificial Intelligence in Warfare Latest Military Technologies in 2025."
140 Malik/Gillani/Anwar, "Military's Guide to Artificial Intelligence," p. 25.
141 Ibid.

- **Cyber Defence and Information Warfare**
  Pakistan is using AI in this filed to bolster cyber defence capabilities and conduct AI-driven offensive cyber operations. The National Cyber Emergency Response Team (National CERT) is the government entity responsible for safeguarding Pakistan's cyberspace. The goal of CERT is to secure critical infrastructure and sensitive data from the ever-evolving landscape of cyber threats. In 2024, the Network Information Centre Pakistan (NIC) and the CERT signed a Memorandum of Understanding (MoU) to enhance the nation's resilience against cyberattacks, cyber terrorism, and cyber espionage.[142] CERT uses AI models to analyse vast volumes of network traffic and system logs to detect anomalies, malware, zero-day attacks, and phishing attempts in real time. In addition, machine learning algorithms continuously monitor networks to flag suspicious behaviour (e.g., unusual access patterns, lateral movement) that might indicate breaches. AI predicts potential attacks by analysing threat actor behaviour, vulnerabilities, and historical breach data—helping pre-emptively secure critical infrastructure.[143]

In addition to these specific projects, Pakistan's ambition to advance digital sovereignty is also reflected in the launch of dedicated initiatives that help build and grow local capacities. To this purpose, the NASTP has been inaugurated in 2023 near the PAF Base Nur Khan in Rawalpindi. The park's activities emphasize innovation, research, and development in collaboration with Turkish defence companies like Baykar and Turkish Aerospace Industry (TAI), which have established joint R&D centres within PAF NASTP. These centres focus on aerospace and cyber technologies, supporting the growth of Pakistan's defence and technology sectors. PAF NASTP is also set to attract foreign investment, create new job opportunities, and foster the development of high-tech industries, with a strong focus on creating state-of-the-art design and R&D centres.[144]

142 "National Emergency Response Team (CERT) and Network Information Center Pakistan Sign MoU to Strengthen Pakistan's Cybersecurity Landscape."
143 Interview with defence expert on emerging technologies, 26 June 2025.
144 "PAF NASTP Inauguration Report – Second to None."

7 Training for Defence AI

As AI-powered systems like drones, surveillance equipment, and decision support tools become standard, military training will need to shift from traditional methods to incorporate AI-enabled platforms. Pakistani initiatives to advance military training for defence AI are emerging gradually. Future training will need to enhance the capabilities of military personnel to operate alongside autonomous systems, interpret AI-driven data, and understand the ethical, strategic, and tactical applications of AI.

In doing so, future training programs will increasingly focus on the integration of machine learning (ML), data analysis, and cybersecurity skills to operate AI systems efficiently. This includes upskilling the workforce to handle AI-enabled technologies in areas like autonomous vehicles, surveillance systems, and AI-guided weapons. Institutes like the National Centre for Artificial Intelligence (NCAI) and PAF's NASTP have already begun laying the groundwork by launching specialized training programs in these fields.

These training programs build in the fact that AI-driven systems can provide instant feedback to trainees, allowing for real-time evaluation of their performance. Whether it is the use of AI-enhanced simulators or autonomous systems, AI will help military instructors monitor performance more accurately, identifying strengths and weaknesses faster than traditional methods. For example, drones like the Shahpar III use AI to assist in precision strikes and surveillance, and these technologies can be incorporated into training to ensure personnel gain expertise in working with such systems.

8 Conclusion

Pakistan's pursuit of defence AI represents a crucial step toward modernizing its military capabilities and addressing the rapidly evolving global defence landscape. The nation's strategic threat perception, particularly concerning India and internal challenges, has driven its adoption of AI to enhance situational awareness, combat effectiveness, cyber defence, and precision targeting. While Pakistan is still in the early stages of integrating AI across its defence sectors, significant strides are being made, notably in the Air Force, where AI is already being implemented in UAVs, air-launched cruise missiles, and surveillance systems.

Pakistan's collaboration with international partners, particularly China and Turkey, has provided the country with the technological expertise and resources necessary to bridge the gap in AI development. The establishment of critical institutions like the National Centre of Artificial Intelligence (NCAI) and the PAF National Aerospace Science and Technology Park (PAF NASTP) underscores the country's commitment to fostering homegrown AI capabilities. These collaborations are not only advancing defence AI but also nurturing innovation within Pakistan's broader technology ecosystem, promising long-term benefits to both defence and civilian sectors.

However, there are significant challenges to overcome. The reliance on foreign technology and expertise, particularly in AI-driven systems, poses risks related to digital sovereignty, data security, and privacy. As AI becomes integral to military operations, there is a need for strong regulatory frameworks to manage these emerging technologies responsibly, balancing national security with ethical considerations. The Pakistani government has already initiated steps in this direction, with the National AI Policy and the National Security Policy acknowledging the role of AI in defence but not yet offering a fully formalized strategy for its deployment.

Internally, the defence sector is undergoing organizational changes to support AI integration, with the Pakistan Army, Navy, and Air Force all taking steps to incorporate AI into their operations. The creation of dedicated AI units, improved interservice coordination, and the establishment of R&D centres are critical to driving innovation and ensuring interoperability with legacy systems. The push for AI in defence is also catalysing a change in human resource management, with a focus on training and developing the necessary skill sets to support this technological shift.

While Pakistan's defence AI capabilities are still evolving, the country is positioning itself as a growing player in this field, committed to maintaining strategic stability and enhancing its deterrence posture. Moving forward, the effective integration of AI into Pakistan's defence strategy will be crucial for its ability to address both current and emerging threats. This process will require continued collaboration with global partners, investment in domestic innovation, and a comprehensive approach to managing the ethical, strategic, and security implications of AI in the military domain.

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